

**Resistant sources of chickpea against dry root rot at
high temperature environment**

THESIS

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By

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2013

CERTIFICATE – I

This is to certify that the thesis entitled, “**Resistant sources of chickpea against dry root rot at high temperature environment**” submitted in partial fulfillment of the requirement for the degree of “**Master of Science in Agriculture**” of the Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur is a record of the bonafide research work carried out by **Mr. Parag Ramesh Khambadkar** under my guidance and supervision. The subject of the thesis has been approved by the Student’s Advisory Committee and the Director of Instructions.

No part of the thesis has been submitted for any other degree or diploma (Certificate awarded etc.) or has been published/published part has been fully acknowledged. All the assistance and help received during the course of investigation have been duly acknowledged by him.

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This is to certify that the thesis entitled **“Resistant sources of chickpea against dry root rot at high temperature environment”** submitted by **Mr. Parag Ramesh Khambadkar** to the Jawaharlal Nehru Krishi Vishwa Vidyalaya, Jabalpur, in partial fulfillment of the requirements for the degree of **“Master of Science in Agriculture”** in the Department of **Plant Pathology** has been, after evaluation, approved by the External Examiner and by the Student’s Advisory Committee after an oral Examination on the same.

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Place : JABALPUR

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(Parag Ramesh Khambadkar)

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ABBREVIATION

DRR	=	Dry root rot
CD	=	Critical difference
CV	=	Coefficient variation
SEm±	=	Standard error mean
SPAD	=	Soil Plant Analysis Development
<i>et al.</i>	=	Co-worker
°C	=	Degree Celsius
psi	=	Per square inch
etc	=	et cetra
eg	=	as for example
BOD incubator	=	Biological oxygen demand incubator
%	=	Per cent
Viz.	=	namely
i.e.	=	that is
No.	=	number
PDA	=	Potato dextrose agar medium
PDB	=	Potato dextrose broth
@	=	at the rate of
LSI	=	Least severity index
HVT	=	Heat tolerant varietal trial
RWC %	=	Relative water content

INTRODUCTION

Gram or Chickpea (*Cicer arietinum* Linnaeus), a member of family Fabaceae, is an ancient self pollinated leguminous crop, diploid annual (2N=16 chromosomes) grown since 7000 BC, in different area of the world (Tekeoglu *et al.*, 2000) but its cultivation is mainly concentrated in semi-arid environments (Saxena, 1990). It is ranked 3rd after common bean (*Phaseolus vulgaris* L.) and pea (*Pisum sativum* L.) and known with different regional names like Dzelbana or Homos (Arabic), Keker (Dutch), Bengal Gram (English), Cafe Franzais (Franch), Kichererbse (German), Garbanzo (Spanish), Cece, Ceci (Roman), Ovetichie harokh (Russian), Shimbira, (Ethiopia) Lablabi, (Turkey) and locally 'chana'.

Chickpea is grown in many tropical, sub-tropical and temperate regions of the world and one of the most important pulse crops due to its multiple functions in the traditional farming system (Sexena and Singh, 1987). Besides helping in the management of soil fertility, particularly in the dry lands, it is an important source of human food and animal feed (Suzuki and Konno, 1982). There are two types of gram, one is the 'Kabuli' white and other is 'desi' brown. Kabuli type is grown in temperate regions while the 'desi' type chickpea is grown in the semi-arid tropics (Malhotra *et al.*, 1987; Muehlbauer and Singh, 1987). In India Chickpea is grown in 8.57 m ha with 7.8 m. tones production and 900-920 kg/ha productivity. This is very low as compared to many other chickpea growing countries like Canada (1270 kg/hect), Mexico (1440), Ethiopia (1191), Nepal (827), Myanmar (1000), Turkey (989) and Australia (1091), (According to the statistics published by FAO, (2006).

The first report of occurrence of root rot in chickpea alongwith wilt was made by Padwick (1948). The species *Rhizoctonia bataticola* responsible for serious diseases. Gupta *et al.* (1983) reported incidence of root rot ranging 3.58 to 20.63 percent in 30 villages of Northern Madhya Pradesh. The disease is more severe at pod formation and grain filling stage especially under stress condition and causes forced maturity resulting reduced seed size.

The Dry root rot (DRR) of chickpea caused by necrotrophic fungus *Rhizoctonia bataticola* (Taub.) Butler is emerging as a serious threat to the chickpea production worldwide (Pande and Sharma, 2010). Dry root rot generally appears during late flowering and podding stages and the infected plants appear completely dried and is an important component of the disease complex that causes root rots and seedling blight in many grain legumes when they are weakened by other stress factors. In the absence of the host crop, it survives in soil as a competitive saprophyte on available dead organic matter.

A critical analysis of the weather data (2000-2010) of the major chickpea growing areas in India indicated higher incidence of DRR in years when temperature exceeds 33°C. Increasing incidence of DRR at various locations over years suggest a strong influence of climate change on disease infection. Savary *et al.* (2011) categorise DRR as an acute-emerging disease that occurs irregularly, both temporally and spatially, may cause massive disruptions in system performances and whose range is expanding to new areas. In chickpea, drying of the plants appears suddenly in the field where DRR affected plants are scattered. Affected plants are usually straw coloured, in some cases the lower leaves and stems show a brown colour. The tap root is black, rotten and devoid of most of the lateral and fine roots. The dead root is quite brittle and shows shredding of the bark. Dark minute sclerotial bodies can be seen on the roots exposed or inside the wood. When the dry stem of the collar region is split vertically, sparse mycelium or minute sclerotia can be seen in the pith. Sclerotial survivability is greatly reduced in wet soils than the dry soils at low moisture levels.

Very little is known about the epidemiology of the disease in relation to predisposition of chickpea plants by climate change variables (temperature and soil moisture). Limited literature on the effect of climatic factors indicates that hot, dry weather promotes infection and development of dry root rot. Chemical control of dry root rot is not effective as *R. bataticola* has a broad host range and survives in soil for longer periods in the form of sclerotia. The sclerotia can survive up to 10 months even in the absence of the host plants and under prevailing dry soil conditions. Use of host plant resistance is the

most economical approach for management of dry root in chickpea. A few chickpea lines with field tolerance to dry root rot have been identified, but high levels of resistance is rare in cultivated genotypes (Anonymous 2010).

The present investigations were undertaken with an objective to identify resistant sources against dry root rot from the available chickpea germplasm lines which are high yielding and heat tolerant. The study will lead to determine the effect of climate change variables (temperature and soil moisture) on DRR infection, colonization and expression under controlled environment besides crop growth stage and histopathological observation as a result of disease initiation leading to the death of the plant. Management through host plant resistance will offers an opportunity in resistance breeding programs. The present investigations were conducted with the following objectives;

Objectives:

1. To find out the dry root rot resistant sources under high temperature environment
2. To know the histopathological alteration and to correlate the dry root rot resistance using relative water content and relative nitrogen content

REVIEW OF LITERATURE

Chickpea is most extensively grown pulse crop in India. It is attacked by air borne, seed borne, and soil borne diseases. (Chattopadhyay *et al.* 2001) More than 75 pathogen have been reported to infect chickpea (Nene *et al.*, 1984).

Among soil borne diseases dry root rot (*Rhizoctonia bataticola* Taub. Butler) is most important diseases causing heavy losses from flowering to podding stage and disease severity depends upon the temperature and moisture condition.

2.1 Historical

The first mention of a plant disease, which may be referred with certainly to *Rhizoctonia* as a causal agent is an important paper by Du Hamel (1728) read before the Paris Academy. In this paper he gave a careful description of a fungus disease of saffron (*Crocus sativus*) occurring in France. He named the fungus *Tuberorides*.

De Candolle (1815) discovered a similar disease on alfalfa. In his taxonomic description he was first person who gave the name for the fungus. Later, after careful study of the characteristics and parasitism of the species, he established the genus *Rhizoctonia* in 1815.

In 1924, Small described a *Rhizoctonia*, which develop black sclerotia of variable size and shape in the diseased roots, and many woody plants from Uganda and named it *Rhizoctonia lamrllifera*. Butler (1925), however considered that the fungus should be regarded as *Rhizoctonia* and made the combination *Rhizoctonia bataticola* (Taub.) Butler to a *Sclerotium* causing a root rot of cotton, french beans and cowpeas.

2.2 Pathogen

The species description of *Rhizoctonia bataticola* is given by Subramaniam (1971). "The basic characteristics of the species are the formation of sclerotia in the host tissue as well as in culture on the infected plant sclerotia seen in the pith region and outer surface of root just below the

bark region. They are brown or jet black, minute, smooth externally, hard, variable in shape, globose, oval oblong, elliptical, curve or even forked and varying in size, but of uniform texture more or less loosely packed.”

The cell of hyphae are barrel shape, branching more or less at right angles and light to dark brown in colour composed of free thick walled cell. The species *Rhizoctonia* is distinguished by light brown to brown hyphae with somewhat loose/ spongy sclerotia in the form of bulbils.

2.3 Occurrence and losses

The first report of the occurrence of root rot in chickpea along with wilt was made by Padwick (1948). The incidence of root rot ranging from 3.58 to 20.63 per cent in 30 villages of northern Madhya Pradesh was reported by Gupta *et al.*, (1983).

The *Rhizoctonia bataticola* responsible for serious diseases like stem blight, foliage blight, root rot, seedling blight, charcoal rot, wilt, collar rot, dry rot, pod and seed rot (Raut and Ingul, 1989) in several crops such as soybean, gram mung, urid, fruit trees and many vegetables (Mukharji and Bhasin, 1986). This *sclerotial* fungus is of wide geographical distribution ranging from Formosa, Philippines, India, Ceylon, East Africa, Palestine and Egypt to the West Indies and the Eastern United State (Ashby, 1927). Significant reduction (25-70%) in chickpea production is caused by this disease in India (Ahmed *et al.*, 1986 and Pandey and Singh, 1990).

Nema and khare (1973) reported that chickpea cultivation in Madhya Pradesh was greatly threatened by soil borne disease generally appearing at two phase of growth early seedling stage and flowering stage wilting was due to *Fusarium oxysporium* f. sp. ciceri, Dry root rot caused by *Rhizoctonia bataticola* and *Sclerotium rolfsii* resulted in collar rot.

Tripathi and Sharma (1983) studied the incidence of chickpea dry root rot in Southern Haryana and reported that the incidence was high from late October to mid November which was decreased in December and January and then again increased in the months of February and March. They also noted that seed sown at a depth of 15 cm developed a less root rot than that sown at 7.5 cm.

Shrivastava *et al.* (2002) conducted a survey in vindhyan planteus zone of Madhya Pradesh India to evaluate the prevalence of wilt and dry root rot disease in chickpea. They reported that *Fusarium oxysporium* was most prevalent (56%) in wilted plant followed by *Rhizoctonia bataticola* (17%) and *Trichoderma sp.* (10%). The incidence of *Rhizoctonia bataticola* also caused withering and drying of plant, which range from 8 to 20 percent.

Studies conducted by Jagre (2012) on succession of chickpea diseases stated that highest percent mortality due to dry root rot was observed in the 10th standard week i.e. 2nd week of march 2012 when the temperature range from 11.6-30.8 0c with 18-76 per cent relative humidity however the disease initiated when the temperature was > 27 0c with the decreases in relative humidity.

2.4 Symptoms

Rhizoctonia bataticola is a soil borne fungus/pathogen. Several workers have reported various types of symptoms produced on plants with varying intensity of the disease in different crops and different regions. Narsimhan (1929) observed that plants infected with *R. bataticola* initially showed drooping of leaves followed by wilting. The main roots of such plants show blackening of finer roots.

According to Mehrotra and Aneja (1990) the first symptom of the disease is yellowing of the leaves within a day or two such leaves drop and in the course of the next two or three days whole plant dies off. The leaves and stem of the affected plants are usually straw coloured. The taproot is dark and quite brittle in dry soil.

Singh *et al.* (1990) observed the root inoculated with *Rhizoctonia bataticola* (Madhya Pradesh) and reported disintegration of cortical tissue while mycelium and sclerotial bodies plugging the xylem were seen in longitudinal section.

2.5 Pathogenicity

Byadgi and Hedge (1985) studied the variation in size and shape of pycnidial and pycnidiospore of *Macrophomina phaseolina* on chickpea based on morphological, cultural character and pathogenic behavior.

Demiro *et al.*, (1994) observed that dry root rot disease of gram is also becoming a serious disease affecting productivity of this crop. They isolated *Rhizoctonia bataticola* from plants and seeds of chickpea cultivars Azizive 94 and tested pathogenicity.

Katariya *et al.* (2012) made studies to determine the cultural and pathogenic variation of *Macrophomina phaseolina* isolates collected from infected chickpea from various parts of Rajasthan. A total of 9 *R. bataticola* isolates, 7 from different districts of Rajasthan and one each from CCS, Haryana Agricultural University, Hisar and Indian Agricultural Research Institute, New Delhi, were isolated from infected chickpea samples. Pure cultures of these isolates were obtained and the pathogenicity of these isolates was tested on 40-day-old chickpea cv. C-235 grown in pots.

2.6 Host plant resistance

Ahmed *et. al.*, (1990) screened 211 promising ICRISAT lines against wilt and root rots mainly (*Rhizoctonia bataticola*), each entry and the susceptible check were grown in 4 m long rows, 30 cm apart in 1987 and 1988 and wilt/root rot incidence was recored at 15 day intervals. The mortality of the susceptible control in 1986-87 main season, 1987 off season, 1987-88 and 1988-89 was 50%, 80%, 95% and 95%, respectively, indicating the uniformity of inoculum. Isolations from dead plants indicated that mortality was mainly due to wilt and dry rot. Among 48 entries which showed less than 20% mortality in the 1987-88 season, 46 lines again showed less than 20% mortality in the 1988-89 season; 22 lines, of which 17 showed less than 10% mortality, were selected for an initial yield trial in 1989-90.

Bakr and Ahmed (1991) reported additional sources of resistance to wilt and root rot of chickpea in Bangladesh, 12 were finally selected as resistance. Of these, ICC 12263 was most resistant, with a mean mortality

rate of 2.3 percent, with other entries exhibiting mean mortality rates of up to 5 percent.

Gangwar *et al.*, (2002) screened 35 chickpea cultivars for resistance to dry root rot caused by *Rhizoctonia bataticola* in a field experiment. All the cultivars exhibited disease symptoms, only 10 cultivars (ICC 2644, 10384, 10630, 112244, 11332, ICCL 81002, 810810, ICC 12263, 12441 and ICCV 90254) were resistant whereas 5 genotypes (GCP 9504, Phule G 96020, 96105, 96313 and GL 91059) were moderately resistant to dry root rot.

Pande, *et al.* (2004) screened 29 chickpea germplasm accessions and 10 cultivars for resistance to dry root rot to confirm their resistance. Artificially inoculated 8-day-old seedlings were incubated at 35⁰ C with 12 h photoperiod. The dry root rot was scored on a 1-9 rating scale on the 8th day after inoculation. Based on the disease score (DS), the accessions were grouped as immune (DS=1), resistant (DS>1 and <=3), moderately resistant (DS>3 and <=5), susceptible (DS>5 and <=7) and highly susceptible (DS>7). Of the 47 lines, none was immune to dry root rot. One germplasm accession (ICC 14395), a cultivar (ICCV 2) and an advanced breeding line (ICCX830203-BH-BH-11H) were resistant to dry root rot. Of the remaining lines, 22 were moderately resistant, 19 susceptible and 3 highly susceptible. The disease severity in the 2 susceptible lines (BG 212 and ICC 12267) used as the controls was rated 9.

Gupta and Babbar (2006) screened 233 *desi* and *kabuli* chickpea genotypes along with 38 promising lines against dry root rot in multiple disease sick plot in Jabalpur, (M.P.) during 2001-03. Genotypes H 99-264, PG 9425-5 and PG 9425-9 (*desi*) and HK 00297 and PG 97-313 (*kabuli*) exhibited resistance against all the 3 pathogens. WCG 97-29, WCG 97-37, IPC 97-72 and H 00-256 (*desi*) showed less than 10% mortality for collar and dry root rots. GF 88240, MPJG 98-9032 and MPJG 98-11551 (*desi*) and HK 94-134 (*kabuli*) showed resistance to wilt and dry root rot diseases. These genotypes are valuable sources of resistance and may be exploited for utilization in breeding programmes for developing high-yielding cultivars with multiple disease resistance.

Vijay *et al.* (2006) evaluated twelve chickpea cultivars for resistance against dry root rot disease under natural and artificial conditions. Screenings under natural conditions were conducted during the rabi seasons of 2001-02 and 2002-03 in Jharkhand, India. For artificial conditions, potted soils were inoculated with mycelial bits of the pathogen and seeds were sown at 4 seeds per pot three days after soil inoculation. Under natural conditions, the mean disease incidence was lowest (8.0%) in G-543, followed by BG-360 and GNG-1365 which recorded mean disease incidence of 9.6 and 9.8%, respectively. These cultivars showed resistant reaction against dry root rot disease. Pusa-372, RSG-888 and RSG-931 showed moderately resistant reaction against the disease. Under artificial conditions, all the cultivars were infected. The mean disease incidence ranged between 14.3 and 41.8%. Cultivars G-543, BG-360 and GNG-1365 showed moderately resistant reaction. Pusa-372, GNG-1292, RSG-888, RSG-895, H-208, CSJ-103 and RSG-931 showed susceptible reaction against the disease. G-543, BG-360 and GNG-1365 showed resistant reaction under natural conditions and moderately resistant reaction under artificial conditions.

Jayalakshmi *et al.* (2008) conducted a pot culture experiment for screening of chickpea entries (12+1 susceptible check JG-62) during *rabi* season of 2004-2005 and 2005-2006. Out of 12 genotypes tested against root rot diseases 4 were found resistant, 2 were moderately susceptible, 3 were susceptible and 3 were highly susceptible to dry root rot disease.

Gupta *et al.* (2012) screened for resistance against *Rhizoctonia bataticola* causing dry root rot in chickpea. The susceptible cultivar BG 212 showed 100 per cent mortality *i.e.* rating 9 in 1-9 scale. The studies led to conclusion that out of 170 accessions, 68 genotypes exhibited resistant reaction (<10% mortality), out of which 26 are the promising lines namely JG1-14, 2-125, 2-4-110, 14-11, 14-10, 2001-13, 2001-13, 2001-18, 2001-80, 2001-115, 2002-20, 2003-95, 2003-14-16, 2004-110, 210, 9605, 1-9, 99-115, 2001-04, 2003-14-2, JG 2000-07, JSC 37, MPJG89-11551, MPJG 89-9023, CSJ 592 and Rajas from JNKVV, Jabalpur. These lines further evaluated for their performance in sick field for three consecutive years and revealed six lines *viz.*, JG 2000-07, JSC 37, MPJG 89-11551, MPJG 89-9023, CSJ 592 and

Rajas as resistant exhibiting <10 per cent mortality, however 14 lines showed moderately resistance reaction.

Indu *et al.* (2012) screened 25 genotypes of chickpea including three checks under natural epiphytotic conditions for their reaction against wilt/root rot complex diseases in two different environments.

2.7 Histopathological

Nene *et al.* (1990) made some histopathological observations of chickpea roots infected by *Rhizoctonia bataticola* which showed dark brown to black discoloration of the pith and xylem vessels in advanced stages of infection in which the healthy noninoculated plant of same age and cultivar were used for comparison. About 1-cm long pieces of root from both *R. bataticola* infected and noninoculated plants were studied for comparison by cutting the transverse section. Upon microscope examination, *R. bataticola* inoculated root showed disintegrated cortical tissues. In additional mycelium and sclerotial bodies of *R. bataticola* plugging xylem vessels were observed in longitudinal sections. This is the first study in which intraxylem mycelium and sclerotial bodies of *R. bataticola* were found in dry root rot infected chickpea and their possible role in sudden death of plants implicated.

Bhatt and Vadhera (1997) studied histopathological studies on cohabitation of *Pratylenchus thornei* and *Rhizoctonia bataticola* on chickpea both singly and jointly (*in vitro*). Both organism caused destruction of epidermal cells; *P. thornei* penetrated the roots directly and *R. bataticola* was thought to secrete macerating enzymes.

Netzahualcoyotl Mayek Perez *et al.* (2002) studied the water relation, histopathology and growth of common bean during pathogenesis of *Macrophomina phaseolina* under drought stress showed higher negative effect than *M. phaseolina* on water relation, vegetative growth and histopathology in *P. vulgaris*.

2.8 Physiological

Yamasaki *et al.* (1999) stated that measurements of Relative water content (RWC %) in leaf tissues are commonly used to assess the water status of plants. Despite its simplicity, this technique needs to be adjusted for

each plant material. Adjustments are mainly related to the length and environmental conditions of the imbibitions period, needed to obtain the turgid mass.

Tejpal *et al.* (2004) conducted an experiment by using six chickpea cultivars (Pusa 256, Pusa 372, BGD 72, DG 36, DG 46 and DG 51) sown on 15 to 30 November or 15 December to determine the effects of temperature as affected by sowing dates on the physiology of the crop. The mean relative water content decreased, whereas the mean membrane injury increased during the vegetative, flowering and pod formation stages of the crop with delay in sowing. The mean total chlorophyll content during the vegetative, flowering and pod formation stages of the crop was highest with 30 November sowing. The mean biological yield decreased, whereas the mean harvest index increased with delay in sowing. The mean seed yield was highest with sowing on 30 November. DG36 was the most superior cultivar tested in terms of the parameters measured.

Van den Berg *et al.* (2004) found that portable, non-destructive chlorophyll meters could be a valuable tool for forest managers and researchers. The ability of such meters to assess both chlorophyll and nitrogen content has been well established for many agricultural species, but not well studied for North American forest tree species.

Studies made by Schlemmer *et al.* (2005) revealed that measuring a crop's physiological parameters provides information for interpreting its response to the environment. Remote sensing is quickly becoming recognized as a valuable tool that has the potential to assess a wide variety of physiological properties over a large area in a short amount of time. This study examined the relationship of corn (*Zea mays* L.) leaf spectral response to its chlorophyll content and relative water content. The effects of N stress and water stress were examined on each of these physiological parameters.

Bahavar *et al.* (2009) conducted the experiment to determine water use efficiency of chickpea (cv. Arman) under water deficit condition and different mineral N application. The experiment was arranged in growth chamber of Mohaghegh Ardabili University, Ardabil, Iran, in 2007, with three nitrogen

levels (25, 50 and 75 kg N ha⁻¹ as urea) and five irrigation regimes (soil water content of 25, 45, 65, 85 and 100% field capacity). Results showed that water stress affect Water Use Efficiency (WUE), Relative Water Content (RWC), Leaf Area (LA), leaf number after stress induction, biomass (above ground tissues) and root/shoot ratio (R/S) traits. Application of Mineral nitrogen increased Water Use Efficiency, RWC, leaf area and biomass. Maximum amount of WUE, RWC, biomass and LA obtained from 75 kg N ha⁻¹ under water deficit condition. This experiment revealed that nitrogen fertilizer could reduce many detrimental effects of water stress. These results also suggest that using of nitrogen fertilizer in water deficit conditions increase tolerance of plants to stress.

Rahbarian *et al.* (2011) studied drought stress effects on photosynthesis, chlorophyll fluorescence and water relations in tolerant and susceptible chickpea genotypes, in order to evaluate morphological and physiological traits related to drought tolerance and to determine the best criteria for screening and identification of drought-tolerant genotypes.

Verma *et al.* (2012) conducted pot experiment during *Rabi* season (2010) to evaluate the effect of brassinolides on physiological traits in chickpea cultivars namely RSG-888 and RSG-895 under drought conditions.

2.9 Resistant sources of chickpea against dry root rot at high temperature environment

Ilyas *et al.* (2000) greenhouse evaluation of 108 chickpea germplasm cultivars/lines for sources of resistance against *M. phaseolina* infection and disease assessment on the basis of external expression of dry root rot symptoms revealed that none of the tested lines was highly resistant. Only one line (97112) was resistant and another line (97104) was moderately resistant. Six lines viz., 97107, 97113, 97119, 97154, 97156 and 97157 exhibited moderately susceptible reaction, while 18 and 81 lines were susceptible and highly susceptible, respectively. When the screened lines were uprooted from pot soil, 150 days after sowing, for observation on root lesions, sclerotial plugging in xylem vessels of roots and collar region, none of the tested lines could be rated as highly resistant, resistant, moderately resistant or moderately susceptible. Only three lines (97104, 97112 and

97144) were rated as susceptible. All the remaining lines were graded as highly susceptible.

Prajapati *et al.* (2004) conducted studies on the interaction of date of sowing with varieties and bio agent on dry root rot of chickpea. Late sowing i.e. 15th nov. significantly reduced the incidence of dry root rot in all three chickpea cultivars.

Prasad *et al.* (2005) studied the effect of weather condition on dry root rot disease in chickpea cv. Pant G 114. The lowest disease incidence (18.1 and 17.3%) of dry root rot and the maximum yield (14.8q/ha and 14.9q/ha) were recorded on 24 Nov. Disease incidence was significantly and positively correlated with maximum, minimum and mean temperature and the minimum and mean relative humidity. The maximum relative humidity was non significantly and negatively correlated with disease incidence. However, rainfall was non significantly and positively correlated with disease incidence. Multiple correlation coefficients between disease incidence and weather variable exhibited strong relationship among the different component and at epidemic during period 2001-2003. Result showed that late sowing significantly reduced the incidence of dry root rot in all three chickpea cultivar. (17.1% in JG-9).

Sharma and Pande (2010) recent studies conducted by researchers at the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) in India indicated that drought caused by prolonged high temperatures make chickpea more vulnerable to dry root rot caused by the fungus *Rhizoctonia bataticola*. The studies indicated that, under emerging climate change, high temperatures and the resultant rapid loss of soil moisture at the reproductive stage of growth predispose chickpea to outbreaks of dry root rot. This is one of the two diseases that affect chickpea production; the other is Fusarium wilt. There are substantial sources of chickpea resistance to Fusarium wilt but none against dry root rot. Studying data taken in India after the rainy season from 2005 to 2010, scientists found that there is higher incidence of dry root rot in chickpea varieties that were resistant to Fusarium wilt in years when temperatures exceed 33°C. Research under different soil moisture and temperature conditions revealed that the

pathogen infected chickpea plants and manifested dry root rot faster at 35°C with soil moisture levels less than or equal to 60%. Fusarium wilt occurs at 25°C and soil moisture levels above 60%. ICRISAT's studies clearly demonstrate the combined roles of drought plus the pathogen that infects chickpea and causes the development of dry root rot.

Sharma and Pande (2013) reported that erratic rainfalls and rise in temperature have become more frequent under the changing scenario of climate particularly in semiarid tropics. As a consequence of it, a drastic shift of chickpea diseases have been recorded throughout the major chickpea growing regions in India and elsewhere. Dry root rot (DRR) caused by *Rhizoctonia bataticola* was found as a potentially emerging constraint to chickpea production than wilt. Increasing incidence of DRR indicate strong influence of climate change variables such as temperature and moisture on the development of disease. The DRR incidence was significantly affected by high temperature and soil moisture deficit. Out of five temperature regimes (15°C, 20°C, 25°C, 30°C and 35°C) and four moisture levels (40%, 60%, 80% and 100%), a combination of high temperature (35°C) and soil moisture content (60%) predisposes chickpea to DRR. The study clearly demonstrates that high temperature coupled with soil moisture deficit is the climate change variables predisposing chickpea to *R. bataticola* infection, colonization and development.

MATERIAL AND METHODS

Present investigations were undertaken to study the **“Resistant sources of chickpea against dry root rot at high temperature environment”**. The materials used and methods followed are described below:

(A) MATERIAL

3.1 General

Borosil make glassware's were used throughout the investigation. These were cleaned with detergent powder followed by rinsing in tap and distilled water. Glasswares were then sterilized in hot air oven at 180°C for 2 hrs. The inoculation needle and other metallic instruments were sterilized by dipping them in alcohol and heating over flame. All precautionary measures were applied to sterilize the inoculation chamber.

The culture media were sterilized in an autoclave at 1.05 kg pressure per square inch (15 lbs psi) for 20 min at 121.6°C. The soil and sand were sterilized at 2.10 kg pressure per square inch (15 lbs psi) for two hours for two consecutive days. The roots of diseased plants were thoroughly washed with tap water and cut into pieces. Root pieces of samples were surface sterilized by dipping them in 1:1000 mercuric chloride solution for 1-2 minutes followed by three washing in sterilized water.

The petriplates containing the media were stored for 24 hrs before use to avoid the possibility of contamination. After isolation the culture was maintained on PDA medium for further investigations.

The Minolta SPAD-502 chlorophyll meter was used, as per Markwell *et al.* (1995), to acquire a rapid estimate of leaf chlorophyll content. The measurements were taken on the upper-most collared leaf halfway from the leaf base to the tip and halfway from the midrib to the leaf margin.

3.2 Test pathogen

Rhizoctonia bataticola, the causal organism of chickpea dry root rot

3.3 Culture Media

The following media were used during the course of investigation and the compositions of media are given below.

(i)	Potato Dextrose Agar medium	-	(PDA)
	Potato (Peeled and sliced)	-	200 g
	Dextrose	-	20 g
	Agar agar	-	20 g
	Distilled water	-	1000 ml
(ii)	Potato Dextrose Broth	-	(PDB)
	Peeled potato	-	200g
	Dextrose	-	20g
	Water	-	1000 ml

3.3.1 Potato Dextrose Agar medium

For potato extraction, peeled and sliced potato was taken and boiled in 500 ml water. Then it was strained through muslin cloth. Agar-agar was washed with water and taken in 500 ml of water then heated until it melted fully. Melted agar agar was poured in potato extract and dextrose was added to it and the solution was finally makeup to one liter. The media was poured in conical flask and sterilized in an autoclave.

3.3.2 Potato Dextrose Broth

Same procedure as described above was employed only agar agar was not added in potato dextrose broth.

3.4 Chemicals

Mercuric chloride solution (1:1000) was prepared and used for surface sterilization of sample during isolation of pathogen and sowing of seed.

(B) METHOD

3.5 Collection, isolation, purification, identification of pathogen, relative water content (RWC%), chlorophyll content (SPAD) and histopathological observation of chickpea infected by pathogen

3.5.1 Collection of diseased plant/root sample

Infected plants showing characteristic symptoms of dry root rot (*Rhizoctonia bataticola*) were collected during the month of December 2012 to March 2013 from chickpea fields, JNKVV, Jabalpur. Samples were brought in to the chickpea pathology laboratory for isolation and further studies.

3.5.2 Isolation of plant sample

The diseased parts of root were cut with the help of sharp razor and rinsed with sterilized water to remove the dirt. The pieces were surface sterilized by dipping in mercuric chloride (1:1000) for 1 min followed by three changes of sterilized water and dried in sterilized blotter paper then transferred aseptically on PDA plates. The plates were incubated at $25\pm 1^{\circ}\text{C}$. The observations were made from the developing colonies. The fungus was sub cultured, purified, identified and maintained on PDA slants.

3.5.3 Identification and purification

Isolation from infected chickpea plants showing symptoms of dry root rot were made on potato dextrose agar medium. *Rhizoctonia bataticola* was found to be associated with root of the diseased plants. The pathogen was identified on the basis of morphological characters with the help of microscope. The purified cultures were maintained on PDA slants.

The culture of *Rhizoctonia bataticola* was purified by sub culturing it through hyphal tip technique respectively and maintained by mass transfer on PDA slants at $25\pm 2^{\circ}\text{C}$ for further study.

3.5.4 Pathogenicity test

Blotter paper technique (Nene *et al.* 1981) was employed for proving the pathogenicity test. The observations were recorded on per cent mortality due to dry root rot and the associated pathogen was isolated.

3.5.5 Screening of chickpea genotypes against dry root rot through Blotter paper technique (*in vitro*)

Five mm disc of the culture was placed on PDA poured petriplates and incubated at 25°C for 4 days. Five mm disc was cut from the above culture and transferred to 250 ml flasks each containing 100 ml P.D. broth. After 5 days of incubation at 25°C two mycelial mats were removed from the flask which were added to 100 ml sterilized distilled water in a beaker after its proper crushing for 1 minute in the blender. Surface sterilized seeds of different chickpea genotypes were sown on plastic trays containing sterilized soil + sand (1:1). Five days old seedlings of each genotype were uprooted in such a way so that root system is not disturbed. The root system of these seedlings was properly washed in running water followed by rinsing in sterilized distilled water. The roots of all genotypes (test lines) were dipped in the inoculum kept in a beaker with an up and down movement for about 30 seconds and the excess inoculum was removed by touching the roots to the edge of beaker.

Ten seedlings of the each test line were taken and 10 of these were kept separately on two blotter papers (size 45 cm x 25 cm with one fold). The blotter paper was moistened adequately and the seedlings were kept in such a way so that only cotyledons and roots are covered and the green tops of the seedlings remains outside the blotter paper.

The folded blotter papers were kept one on top of the other, in heaps of ten in a tray. A susceptible check was kept in each heap. These trays were placed into the incubator at 35°C for 8 days. Artificial light was provided for 12 hrs and the blotter papers were moistened adequately on alternate day.

Rating scale

In order to find out the resistant genotype of chickpea, test lines were scored at the end of the incubation period by examining the seedlings for the extent of root damage and were scored for the disease on 1-9 point scale as mentioned below:

Rating scale used for disease scoring

Rating	Category	Symptoms of DRR	DRR Percent
1	Resistant	No infection on roots	0.0-10
2-3	Moderately resistant	Very few small lesions on roots	10.1-20
4-5	Tolerant	Lesions on roots clear but small, new roots free from infection	20.1-30
6-7	Susceptible	Lesions on roots many, new roots generally free from lesions	30.1-40
8-9	Highly susceptible	Roots infected and completely discolored	40.1 and above

DRR – Dry root rot

3.5.6 Screening of chickpea genotypes against dry root rot resistance under multiple disease sick field

In order to identify the sources of resistance, the chickpea lines were screened in disease sick plot which was prepared by adding the culture of *Rhizoctonia bataticola* in chickpea straw and sand medium. The culture was mixed in the soil (1:9) then added in each row before sowing.

The experiment was conducted in 2 replications. For each test line, 40 seeds / entry was sown in 4 m. long row with seed to seed spacing of 10 cm and row to row spacing of 30 cm.

A susceptible check of JG 62 (early wilter) was maintained after every 2 rows of test entries for comparison (Nene *et al.* 1981). A resistant cultivar

JG 315 was also sown in each trial for comparing the level of resistance in the test entries of the trials. Sowing was done on 15th Nov. 2012 after applying recommended dose of fertilizer. One come up irrigation was given to the crop. Observations on per cent emergence were recorded after 20 days of sowing and mortality due to dry root rot were recorded weekly from flowering to podding stage. The crop was harvested in the maturity stage (April, 2013)

Percent mortality due to dry root rot was calculated by using the following formula;

$$\text{Percent mortality} = \frac{\text{Number of diseased plants}}{\text{Total number of seedlings}} \times 100$$

Disease reaction :- Dry root rot

Rating	Category	Reaction type	Percent mortality(%)
1	R	Resistant	0-10
2	MR	Moderately resistant	10.1-20
3	T	Tolerant	20.1-40
4	S	Susceptible	40.1-60
5	HS	Highly susceptible	60.0 and above

3.5.7 Measurement of Relative water content of leaf

Measurements of relative water content (RWC %) were perform on leaves collected from the 5th, 6th, and 7th of leaflet. Leaves were then immediately weighed (fresh weight, FW). The FW obtained from each sample i.e. above 0.5 g. In order to obtain the turgid weight (TW), leaves were floated in distilled water inside a closed petri dish. During the imbibitions period, leaf samples were weighed periodically, after gently wiping the water from the leaf surface. At the end of the imbibition period, leaf samples were placed in a pre-heated oven at 80°C, for 48 h, in order to obtain the dry weight (DW). All weight measurements were made using an analytical scale, Values of FW,

TW, and DW were used to calculate RWC, given by Barrs and weatherley (1962)

$$\text{RWC (\%)} = [(\text{FW} - \text{DW}) / (\text{TW} - \text{DW})] \times 100$$

3.5.8 Measurement of relative nitrogen content

The SPAD–502 a hand held chlorophyll meter (Minolta Corporation) was used for rapid and non-destructive estimation of chlorophyll in leaves. This instrument uses a silicon photo-iodide to detect transmittance of light emitted by two light emitting diodes through a leaf sample, one with peak emittance at 650 nm where absorbance by chlorophylls is high and relative unaffected by carotene and one with peak emittance at 940 nm where absorbance by chlorophyll is negligible. Chlorophyll content was measured in 5th, 6th and 7th leaf of leaflet at flowering and podding stage and expressed as SPAD unit (Soil Plant Analysis Development). Three measurement were taken per plant of each genotypes. The results were than averaged, resulting in a single value to represent that genotype. Leaf optical properties were also acquired at the same location and in the same manner, as was the SPAD meter data.

3.5.9 Histopathological observation of chickpea dry root rot

To study host pathogen relationship, histopathological observations of chick pea roots infected by *Rhizoctonia bataticola* showing dark brown to black discoloration of the root of different entries rated as 1-9 scale and healthy non-inoculated plant of same age of the different cultivars were used for comparison. About 1-cm long pieces of root from both *R. bataticola* infected and non-inoculated plants were fixed in FAA for 24 hrs. Hand cut Transverse section with sharp razor were cut and stained with 3% cotton blue and lactophenol. The slides were examined under microscope to see the entry and interaction of the pathogen in the host tissue.

RESULT

4.1 Collection of diseased samples

Dry root rot infected plants of chickpea were collected from disease sick field located at Seed Breeding Farm, Jawarahlal Nehru Krishi Vishwa Vidyalaya, Jabalpur, during 2013.

4.2 Symptoms

Dry root rot infected chickpea plants were observed in the field at flowering to podding stage. The leaves and stems of affected plants showed straw-coloured appearance. The disease appeared in a severe form when ambient temperature was $>30^{\circ}\text{C}$. The pathogen formed reddish brown sunken lesions on the stem and tap root most frequently near the soil line causing stunting growth or death of the plants.

The roots were usually dry unless the soil is wet. The tap root becomes quite brittle, showed shredding of the bark and broken very easily. Observation of the infected plants revealed the presence of sclerotia in the pith region and outer of the root just below the bark region.

4.3 Isolation, Purification and identification of *Rhizoctonia bataticola*

Isolation from dry root rot affected plants done on PDA medium revealed the association of *Rhizoctonia bataticola* which was identified on the basis of morphological characters. The culture was purified by using the hyphal tip technique and the purified culture was maintained on PDA slants.

4.4 Pathogen

The growth pattern of the colony of *Rhizoctonia bataticola* appears to be somewhat cottony, fluffy and light brown to dark brown or grayish black in colour. The microscopic observations denote that mycelial cells of hyphae are septate, branched and light to dark brown in colour. Sclerotia are brown to jet black and variable in shape i.e. round to oval and irregular.

**Plate-1 : Plant showing symptoms of dry root rot affected by
*Rhizoctonia bataticola***

A. Dry root rot affected chickpea plant

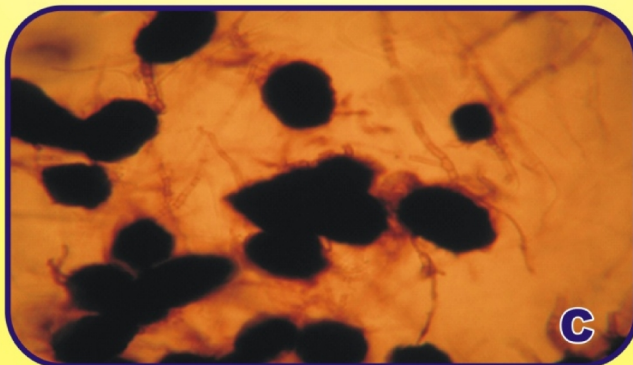
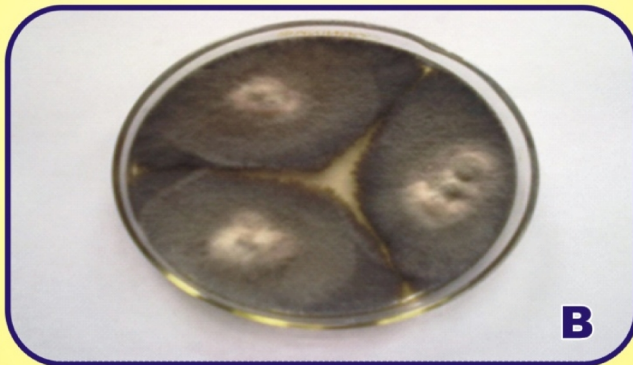
➤ Growth pattern and pigmentation of isolate of *Rhizoctonia bataticola* on Potato dextrose agar medium

B. Front view

➤ Morphological characters of *Rhizoctonia bataticola* as observed in microscope

C. Sclerotia & hyphae

PLATE- 1



4.5 Pathogenicity test

The pathogenicity test was conducted by using susceptible variety BG 212 through Blotter paper technique (Nene et al. 1981). The observations were recorded on percent mortality due to dry root rot and the associated pathogen was isolated. Symptoms on roots appeared after 08 days of incubation at 30⁰c the root exhibited brown to black discolouration, where as the roots were completely healthy having white root system in uninoculated seedling of the same variety.

4.6 Host plant resistance

To know the resistant sources against dry root rot chickpea genotypes which were previously identified as heat tolerant lines as well as germplasm lines were evaluated in sick field and through blotter paper technique and the data is presented in Tables 1, 2, 3, and 4.

Table 1. Screening of Heat tolerant lines for dry root rot resistance (HVT Desi)

Entries	Sick field					Blotter paper technique		
	Emergence		Percent mortality		Average	Reaction	Rating	Reaction
	R1	R2	R1	R2				
ICC 16181	35	25	11.42	4.0	7.7	R	1	R
JG 19	20	27	45.0	40.7	42.8	S	6-7	S
JG 1307	20	33	15.0	4.4	30.2	T	4-5	T
MPJG 99-115	20	20	10.0	50.0	30.0	T	4-5	T
ICC 8474	25	34	8.0	23.5	15.7	M R	4-5	T
JG 18	31	15	12.9	13.3	13.1	M R	2-3	M R
JG 16	24	34	-	8.8	4.4	R	1	R
ICCV 07102	25	35	-	14.2	7.1	R	1	R

Among eight medium heat tolerant entries tested for dry root rot resistance under field as well as artificially inoculated condition through blotter test revealed that three entries viz. ICC 16181, JG 16 and ICCV 07102 exhibited resistant reaction (Rating1) under both field and *in vitro* condition. JG 19 was found susceptible under both the screening technique (Table 1) the disease incidence ranged from 4.4 - 42.8 percent.

Table 2. Screening of Heat tolerant (Large seeded) lines for dry root rot resistance

	Sick field						Blotter paper technique	
Entries	Emergence		Percent mortality		Average	Reaction	Rating	Reaction
	R1	R2	R1	R2				
JG 74 x IPC-4958	24	21	12.5	100	56.2	S	6-7	S
JG 2003 14 16	31	28	9.6	10.7	10.1	M R	4-5	T
ICC 96029 x A11	36	26	33.5	100	66.7	H S	8-9	H S
ICCV 07110	35	17	24.1	23.5	23.8	T	4-5	T
ICCV 07118	34	32	2.9	3.1	3.0	R	2-3	M R
JG 21	36	40	11.1	21.4	16.2	M R	4-5	T
ICCV 07117	34	39	11.7	10.2	10.9	M R	2-3	M R
GG 2	17	22	11.7	40.9	26.3	T	4-5	T
ICCV 07109	18	22	16.6	22.7	19.6	M R	4-5	T
JG 17	33	28	9.0	10.7	9.8	R	2-3	M R
ICC 4958	32	27	25.0	37.0	31.0	T	4-5	T
ICCV 11101	31	28	6.4	17.8	12.1	M R	4-5	T
ICCV 11109	29	35	13.7	11.4	12.5	M R	2-3	M R
JG 130	30	40	23.	22.5	22.9	T	4-5	T

JG 14 x 4958	14	12	0.0	16.6	8.3	R	2-3	M R
JG 130 x ICC 11551	14	15	0.0	6.6	3.3	R	2-3	M R
ICCV 11116	26	30	15.3	100	57.6	S	6-7	S
ICCV 11114	36	34	5.5	2.9	4.2	R	2-3	M R

Of the 18 large size desi chickpea entries evaluated for dry root rot resistance, 5 entries viz. ICCV 07118, JG 17, JG14x4958, JG 130 x ICC11551, ICCV 11114 resulted resistant under field condition but moderately resistant to dry root rot when screened through blotter paper technique. The dry root rot incidence in sick field ranged from 4.2 – 66.7 percent (Table 2).

Table 3. Screening of chickpea genotypes for dry root rot resistance under National nursery trial

	Sick field					Blotter paper technique		
Entries	Emergence		Percent mortality		Average	Reaction	Rating	Reaction
	R1	R2	R1	R2				
GLK 281427	11	13	81.8	86.9	84.3	H S	6-7	S
GNG 1969	15	15	0.0	6.6	3.3	M R	2-3	M R
RG 2003-28	32	35	9.3	8.5	8.9	R	2-3	M R
Vihar	13	10	69.2	30.0	49.6	S	6-7	S
GJG 0906	33	39	3.0	2.5	2.7	R	2-3	M R
SC-RS-1	14	11	28.5	26.2	27.3	T	4-5	T
IPCK 06-70	18	17	12.5	0.0	6.2	R	2-3	M R
JG 24	33	36	6.0	5.5	5.7	R	2-3	M R
JG 14	42	44	7.1	13.6	10.3	M R	2-3	M R
JGK 13	15	17	80.0	5.8	42.2	S	6-7	S
GNG 1488	37	33	16.2	18.1	17.1	M R	4-5	T
BDG 112	41	44	7.2	4.5	5.8	R	2-3	M R
GNG 1996	38	43	5.2	27.9	16.5	M R	4-5	T
BG 3004	36	36	2.7	5.5	4.1	R	2-3	M R

This trial comprised of 14 entries of which 8 entries viz. GNG 1969, RG 2003-28, GJG 0906, IPCK 06-70, JG 24, JG14, BDG 112, BG 3004 were found resistant to dry root rot having 3.3 – 8.9 percent disease incidence. Further the entries exhibiting resistant reaction under field condition exhibited moderately resistant under blotter paper technique. (Table 3)

Table 4. Confirmation of chickpea entries for dry root rot resistance

	Sick field					Blotter paper technique		
Entries	Emergence		Percent mortality		Average	Reaction	Rating	Reaction
	R1	R2	R1	R2				
GNG 1488	31	46	6.4	30.1	18.2	M R	2-3	M R
IPC 05-28	48	48	4.1	18.7	11.4	M R	1	R
H 04-31	43	46	6.9	4.4	5.6	R	2-3	M R
GNG 1996	45	38	20.0	57.8	38.2	T	4-5	T
IPCK 06-78	14	21	7.1	4.7	5.9	R	1	R
IPCK 113	22	24	13.6	8.3	10.9	M R	2-3	M R
BCP 49	18	28	16.6	10.7	13.6	M R	4-5	T
BCP 26	24	18	25.6	22.4	23.6	T	6-7	S
BCP 2010-1	24	14	4.1	21.4	12.6	M R	2-3	M R
BCP 60	27	15	11.1	20.0	15.5	M R	2-3	M R

The data presented in table 4 indicate that among the 10 entries screened for disease resistance 2 entries i.e. H 04-31 and IPCK 06-78 exhibited resistance to dry root rot however, 5 entries (GNG 1488, IPC 05-28, IPCK 113, BCP 2010-1, BCP 60) were found to be moderately resistant. The disease incidence under sick field ranged from 5.6 – 38.2 percent. The entries were either resistant / tolerant / moderately resistant under both the screening methods.

PLATE- 2



Plate 2. Screening of chickpea genotypes for dry root rot resistance under national nursery trial

4.7 Histopathological alteration

Infected plant or seedling showed the typical symptoms such as deep or irregular necrotic lesions they were extended across root surface. The fungus adhered to the roots and then produced the infected hyphae, *Rhizoctonia bataticola* developed faster under water stress as compared to irrigated condition. Susceptible cultivars showed typical symptoms after emergence while resistant showed small lesions in the epidermal and external cortex cell.

On examination of hand cut transverse section taken from inoculated seedling revealed that microsclerotia comes in contact with the chickpea surface, developed hyphae and penetrate in between epidermal cell and then the fungus grow intercellularly. Afterwards the fungus caused cellular collapse, necrosis of epidermal and cortex cell in roots of chickpea and tissues degeneration which accelerated the death of host. The damage cell showed disintegration of the vascular system and significant disorganization of pith parenchyma and also disintegration of xylem and phloem parenchyma and pith cells was detected. (Plate 2) Transverse section from the root of resistant plant did not showed blackening of xylem vessel, epidermal cell and cortex cells. However, only in some cases the pathogen entry was found in traces either in epidermal cells / pith cells. (Plate 2) Similar transverse section of infected root showing different rating scale exhibited the degree of establishment of the host pathogen interaction.

Plate - 3. Transverse sections of chickpea roots, subjected to high temperature environment and *Rhizoctonia bataticola* inoculation

- A.** Showing healthy tissue of chickpea root
- B.** Showing cellular collapse and necrosis of cortex parenchyma
- C.** Showing cellular necrosis of epidermis
- D.** Showing microsclerotia initiating pathogenesis
- E.** Showing necrosis of cortex
- F.** Showing pith disintegration

PLATE- 3

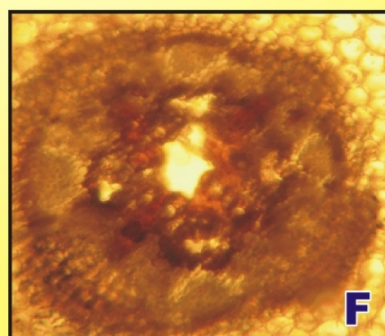
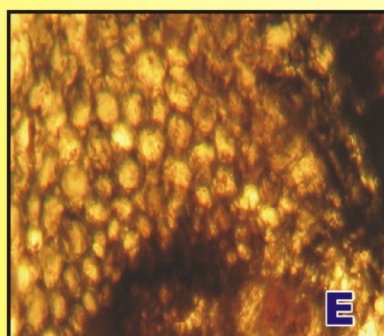
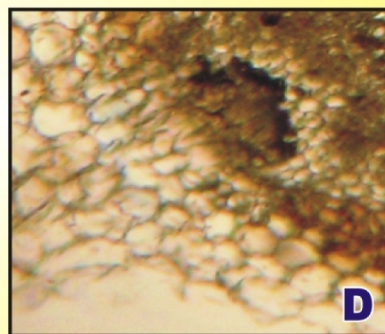
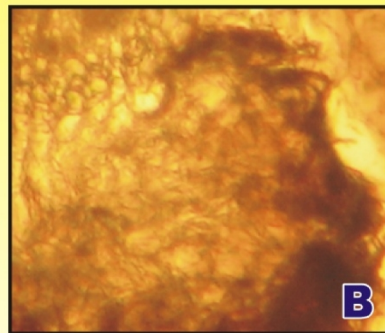
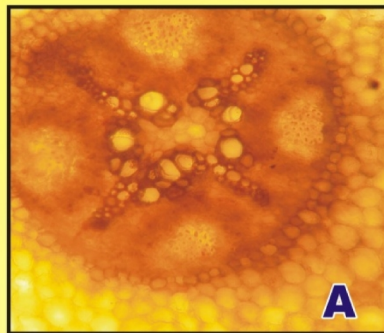
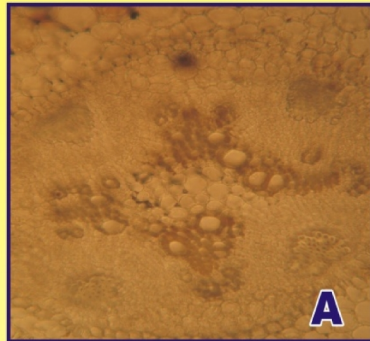


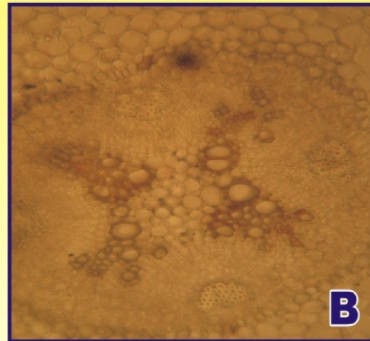
Plate – 4. Transverse section of chickpea root (Uninoculated and inoculated) showing host pathogen interaction

- A.** Uninoculated healthy tissue of root
- B.** Resistant - having no infection on the tissue of root
- C.** Moderately resistant - where xylem vessels infected in traces
- D.** Tolerant - showing disintegration of xylem and phloem vessels
- E.** Susceptible - sclerotia present all over
- F.** Highly susceptible - tissue where root completely discolored

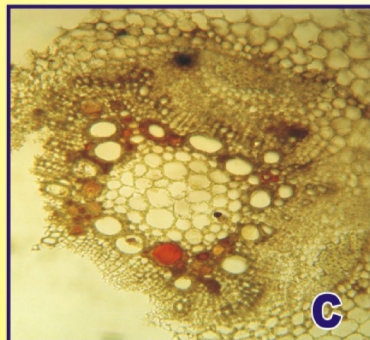
PLATE- 4



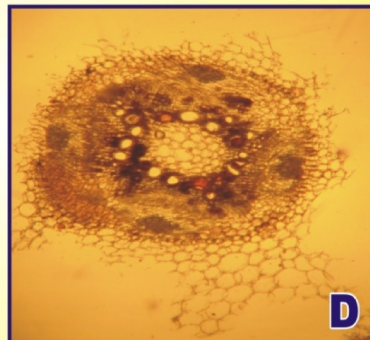
A Uninoculated root



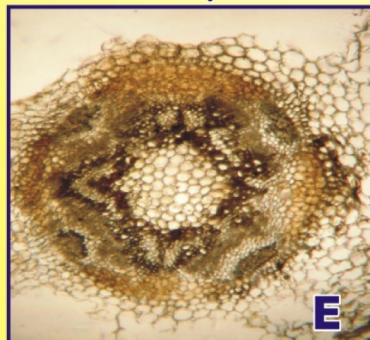
B Resistant



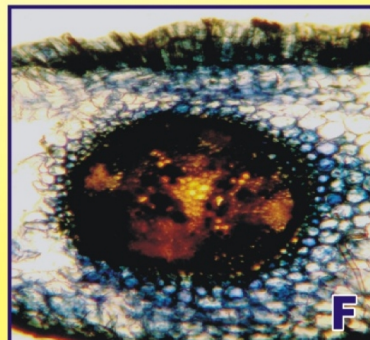
C Moderately resistant



D Tolerant



E Susceptible



F Highly susceptible

4.8 Physiological correlation

Influence of Relative water content (RWC %), Chlorophyll content (SPAD Unit) on resistance / susceptibility of chickpea genotypes against dry root rot

To know the physiological correlation and its response in making the chickpea plants resistance or susceptible to dry root, the data on relative water content (RWC %), and chlorophyll content (SPAD) was compared with the percent mortality occurred in sick field.

The data presented in (station trial) heat tolerant varietal trial (HVT Desi) showed that resistant genotypes had maximum relative water content or water holding capacity in leaves as in JG 16 (68.6%) followed by ICC 16181 (66.7%). Maximum chlorophyll content (SPAD unit) was also noted in JG 16 (64.1), and ICC 16181 (55.5) respectively. Both the genotypes showed the minimum percent mortality i.e. 4.4 and 7.7 in sick field respectively (Table 5 and fig 1).

Similar observations were recorded in HVT (Large Desi) trial where 18 entries were screened. The data exhibits that maximum relative water content was noted in ICCV 07118 (74.6%), followed by JG 14 x 4958(70.5%), JG 130 x ICC 11551(77.0%), and ICCV 11114(73.8%) and maximum chlorophyll content (SPAD) was noted in ICCV 07118 (68.5), JG 14 x 4958 (71.8), JG 130 x ICC 11551 (67.3), and ICCV 11114(60.4). These genotypes having very low percent mortality i.e. < 10% and were rated as resistant to dry root rot (Table 6 and fig 2). Whereas, susceptible genotypes possess less relative water content (RWC %) and chlorophyll content as compared to resistant genotypes.

In National nursery dry root rot, maximum relative water content was noted in IPCK 06-70(71.7%) followed by BDG 112(63.2%) and BG 3004(64.9%) and also maximum chlorophyll content (SPAD) was noted in IPCK 06-70 (70.4), BDG 112(61.2), BG 3004(65.5) respectively. Percent mortality in these genotypes have been observed < 10 thus graded as resistant (Table 7 and fig. 3).

In another trial i.e. Confirmation dry root rot resistant entries; H 04-31 (66.6%) and IPCK 06-78(73.8%) possess higher value of relative water content as well as chlorophyll content (SPAD) i.e. 70.4 and 60.2 respectively. Percent mortality was minimum < 10 and hence graded as resistant genotypes (Table 8 and fig. 4).

Table 5. Comparative performance of heat tolerant lines (Desi) for their Relative water content (RWC %), chlorophyll content and percent mortality due to dry root rot

S. No.	Entries/Genotype	RWC%	Chlorophyll Content (SPAD)	Percent mortality
1	ICC-16181	66.7	55.5	7.7
2	JG-19	54.6	40.4	42.8
3	JG-1307	60.2	27.3	30.2
4	MPJG-99-115	66.7	27.4	30.0
5	ICC-8474	67.2	58.8	21.7
6	JG-18	59.3	59.3	13.1
7	JG-16	68.6	64.1	4.4
8	ICCV-07102	59.4	52.4	7.1
	Mean	62.84	48.15	LSI = 19.63
	SEm±	1.88	3.21	
	CD at 5%	5.69	9.74	
	CV (%)	5.17	11.55	

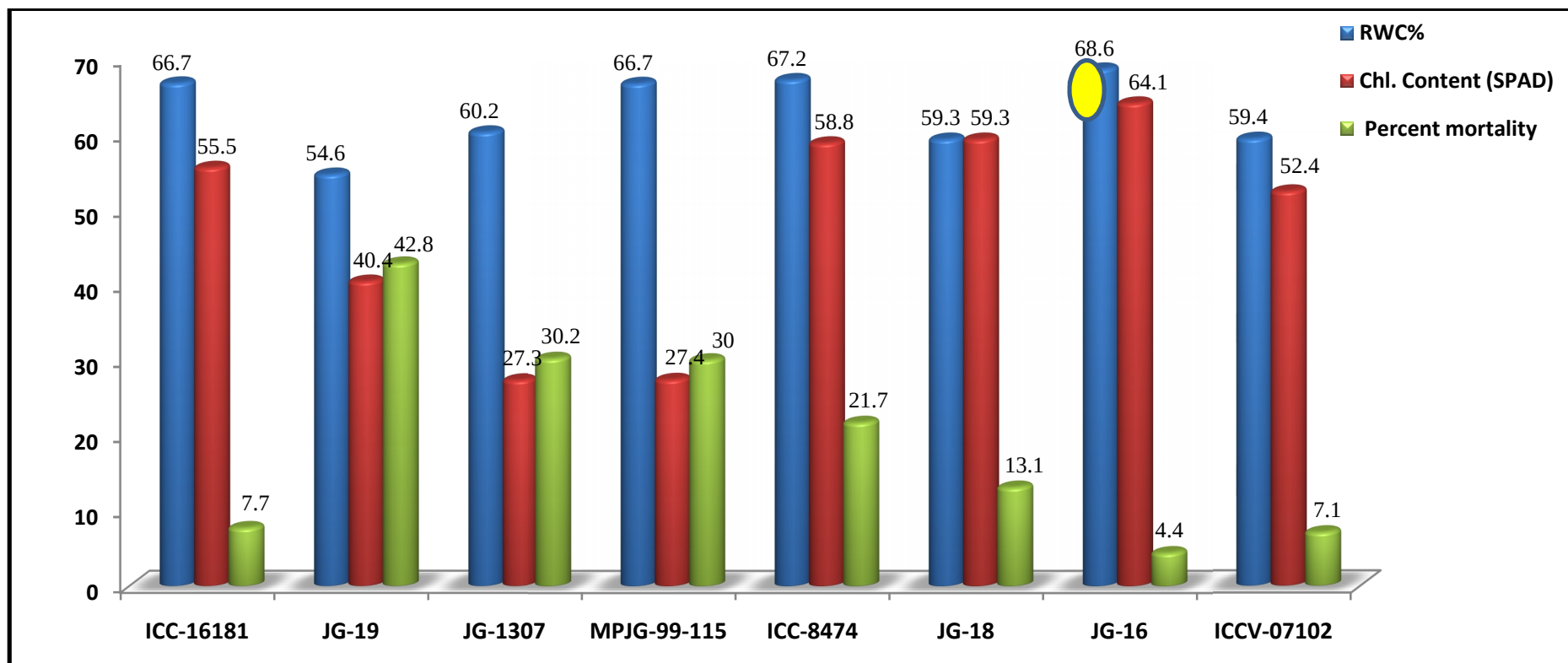


Fig. 1 Comparative performance of heat tolerant lines (Desi) for their Relative water content (RWC %), Chlorophyll Content (SPAD) and percent mortality due to dry root rot

Table – 6. Comparative performance of heat tolerant lines (Large seeded) for their Relative water content (RWC %), Chlorophyll Content and percent mortality due to dry root rot

S. No.	Entries/Genotype	RWC%	Chlorophyll Content (SPAD)	Percent mortality
1	JG-74*IPC-4958	67.9	37.8	56.2
2	JG-2003-14-16	68.4	30.2	10.1
3	ICC-96029*A11	56.0	40.0	66.7
4	ICCV-07110	65.0	55.0	23.8
5	ICCV-07118	74.6	68.5	3.0
6	JG-21	63.5	49.1	16.2
7	ICCV-07117	66.2	44.6	10.9
8	GG-2	66.4	58.5	26.3
9	ICCV-07109	64.9	62.7	19.6
10	JG-17	68.4	34.6	9.8
11	ICC-4958	62.2	53.5	31.0
12	ICCV-11101	64.9	57.3	12.1
13	ICCV-11109	62.2	55.6	12.5
14	JG-130	63.3	66.4	22.9
15	JG-14*4988	70.5	71.8	8.3
16	JG130*ICC11551	77.0	67.5	3.3
17	ICCV-11116	57.8	39.2	57.6
18	ICCV-11114	73.8	60.4	4.2
	Mean	66.28	52.93	LSI = 25.39
	SEm±	0.99	3.09	
	CD at 5%	2.86	8.64	
	CV(%)	2.59	9.83	

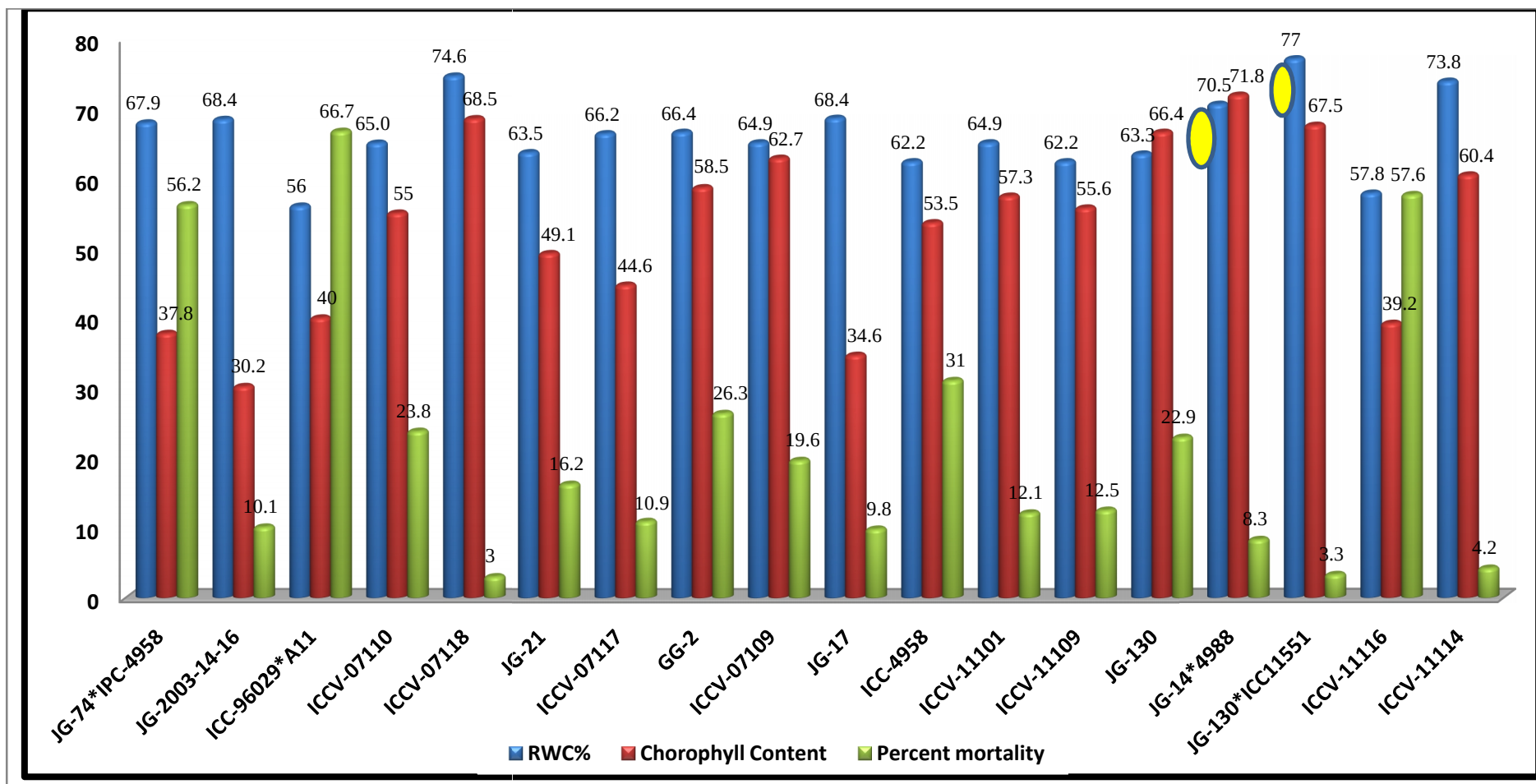


Fig. 2 Comparative performance of heat tolerant lines (Large seeded) for their Relative water content (RWC %), chlorophyll content (SPAD) and percent mortality due to dry root rot

Table 7.: Comparative performance of chickpea genotypes (National nursery) for their Relative water content (RWC %), Chlorophyll Content and percent mortality due to dry root rot

S.No.	Entries/Genotypes	RWC(%)	Chlorophyll Content (SPAD)	Percent mortality
1	GLK 281427	64.6	60	84.3
2	GNG 1969	60.6	50.5	3.3
3	RG 2003-28	63.2	57	8.9
4	Vihar	60.5	49.1	48.1
5	GJG 0906	60.4	55.2	2.7
6	SC-RS-1	57	42.4	25.6
7	IPCK 06-70	71.7	70.4	6.2
8	JG 24	61.4	58.7	5.7
9	JG 14	60.4	57.6	10.3
10	JGK 13	51.8	49.9	42.2
11	GNG 1488	59.9	51.3	17.1
12	BDG 112	63.2	61.2	5.8
13	GNG 1996	56.7	59.6	16.5
14	BG 3004	64.9	65.5	4.1
	Mean	61.16	56.31	LSI = 21.18
	SEm±	0.84	1.71	
	CD at 5%	2.45	4.9	
	CV(%)	2.38	5.27	

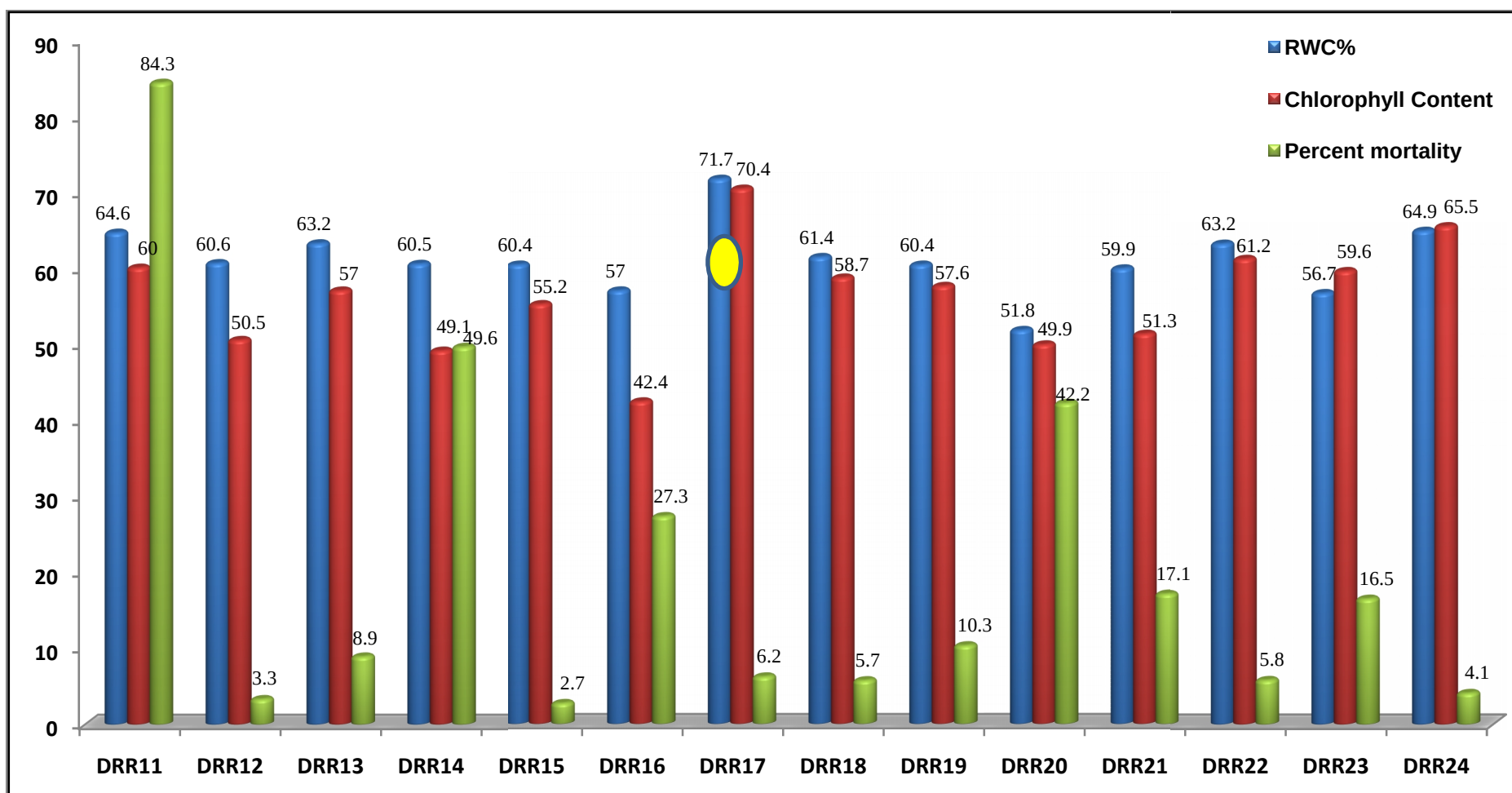


Fig. 3 Comparative performance of chickpea genotypes (National nursery) for their Relative water content (RWC %), Chlorophyll Content (SPAD) and percent mortality due to dry root rot

**Table - 8. Comparative performance of chickpea genotypes
(Confirmation trial) for dry root rot resistance and
physiological parameter**

S. No.	Entries/Genotypes	RWC (%)	Chlorophyll Content (SPAD)	Percent mortality
1	GNG 1488	59.5	52.8	18.2
2	IPC 05-28	62.1	66.7	11.4
3	H 04-31	66.6	70.4	5.6
4	GNG 1996	59.4	50.4	38.2
5	IPCK 06-78	73.8	60.2	5.9
6	IPCK 113	65.2	51.3	10.9
7	BCP 49	60.9	55.7	13.6
8	BCP 26	54.3	51.3	23.6
9	BCP 2010-1	64.1	66.6	12.6
10	BCP 60	60.3	64.1	15.5
	Mean	62.62	58.95	LSI = 15.55
	SEm±	0.81	6.68	
	CD at 5%	2.42	Non significant	
	CV(%)	2.25	19.2	

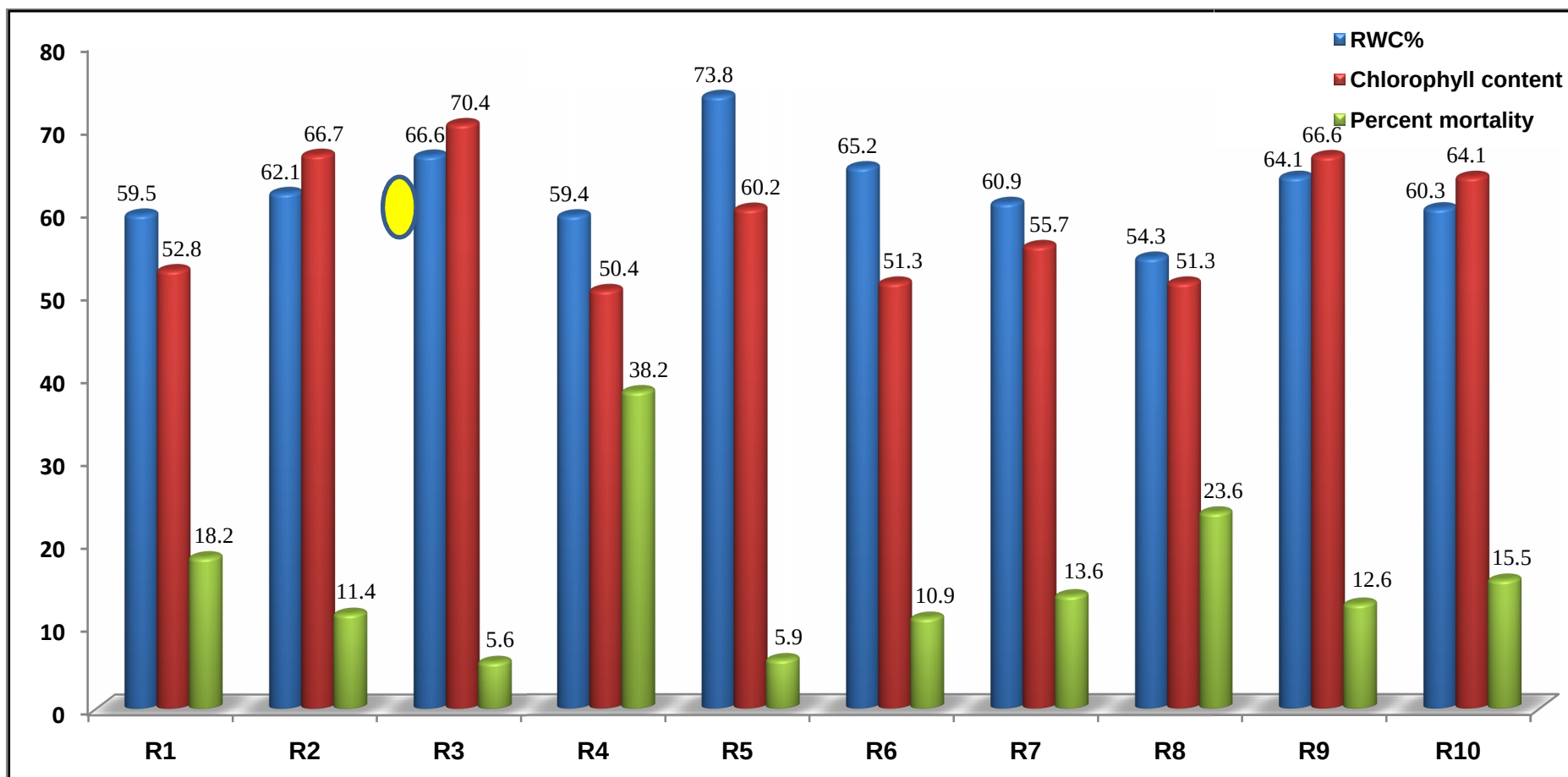


Fig. 4 Comparative performance of chickpea genotypes (Confirmation trial) for dry root rot resistance and physiological parameter

DISCUSSION

The present investigations on “Resistant sources of chickpea against dry root rot at high temperature environment” were undertaken with an objective to identify the resistant sources in the available germplasm and its relation with regards to relative water content and chlorophyll content and its role in making the plant resistance.

Dry root rot of chickpea caused by *Rhizoctonia bataticola* a soil borne pathogen is the most important disease and cause enormous losses at podding and maturity stage when the temperature exceeds $> 30^{\circ}$ C. Several workers have reported various types of symptoms produced on plants with varying intensity of the disease in different crops and different regions. Narsimhan (1929) observed that plants infected with *Rhizoctonia bataticola* initially showed drooping of leaves followed by wilting. The main roots of such plants show blackening of finer roots.

The disease was found to appear in post flowering and pod formation stage. Nema and Khare (1973) reported that chickpea cultivation in Madhya Pradesh was greatly threatened by soil borne disease. Gupta *et al.* (1983) reported the incidence of root rot ranging from 3.2 to 20.6 per cent in 30 villages of northern Madhya Pradesh. Nene *et al.* (1984) also recorded heavy losses to chickpea crop due to dry root rot caused by *Rhizoctonia bataticola*.

Dry root rot symptoms observed in diseased plants exhibited straw colour appearance and tap root mainly becomes brittle having minute sclerotia. These symptoms separated this disease from the so called “Wilt complex” as stated by various workers in India (Nene *et al.* 1989). Maximum atmospheric temperature up to 35° C favoured the disease development. Substantial mortality and losses in chickpea due to *Rhizoctonia bataticola* had been reported in high ambient temperature 33° C and above in the post flowering stage (Singh and Mehrotra 1982).

Chickpea entries / genotypes were evaluated to find out the host plant resistance through Blotter paper technique. The genotypes evaluated for host resistance have been graded as resistant, moderately resistance, tolerant, susceptible and highly susceptible (1-9 point rating scale). Out of 50 genotypes 5 entries namely ICC 16181, JG 16, ICCV 07102, IPC 05-28 and IPCK 06-78 having resistant against dry root rot have been identified. There was no blackening of the root system as it was quite healthy, completely developed and exhibited white appearance and showed < 10 per cent mortality. Furthermore, 21 entries have been observed in the range of 10-20 percent and graded as moderately resistant. Chickpea genotypes have been screened and identified as resistant to dry root rot by various workers on the basis of *in vitro* as well as field evaluation test screening (Nene *et al.*, 1981, Ilyas & Khalid, 2000; Gupta *et al.*, 2003, Gupta *et al.*, 2012)

Fifty entries / genotypes screened in the infested sick plot revealed the disease incidence in the range of 2.7 to 100 percent. Seventeen entries namely ICC 16181, JG 16, ICCV 07102, ICCV 07118, JG 17, JG 14 X 4958, JG 130 X ICC11551, ICCV 11114, GNG 1969, RG 2003-28, GJG 0906, IPCK 06-70, JG 24, BDG 112, BG 3004, H 04-31 and IPCK 06-78 exhibited < 10 percent mortality and graded as resistant and 17 entries namely ICC 8474, JG 18, JG 2003 14 16, JG 21, ICCV 07117, ICCV 07109, ICCV 11101, ICCV 11109, JG 14, GNG 1488, GNG 1996, GNG 1488, IPC 05-28, IPCK 113, BCP 49, BCP 2010-1, BCP 60 exhibited 10.0 to 20.0 percent mortality and graded as moderately resistant. Large number of chickpea cultivars have been screened and identified resistant genotypes by various workers (Nene *et al.* 1989; Reddy *et al.* 1990; Bhatt 1993; Gupta, 1997; Gangwar *et al.* 2002; Pandey *et al.* 2004; Gupta and Babbar, 2006, Gupta *et al.*, 2012). Consistant resistant genotype under multilocation evaluation test is not available. However, Gupta et al. (2012) have identified 6 lines as resistant to dry root rot.

In histopathological observation *Rhizoctonia bataticola* inoculated root showed disintegrated cortical tissues. In additional mycelium and sclerotial bodies of *R. bataticola* plugging xylem vessels were observed in transverse sections. Showing the intraxylem mycelium and sclerotial bodies of *R. bataticola* in dry root rot infected chickpea and their possible role in sudden death of plants. These studies are in confirmation to the findings of Nene *et al.* (1990) who reported dark brown to black discoloration of the pith and xylem vessels in advanced stages of infection in which the healthy uninoculated and inoculated plant were used for showing host pathogen interaction.

Susceptible cultivars showed typical symptoms in the epidermal and external cortex cell. Where sclerotia comes in contact with the chickpea surface developed hyphae and penetrate in between epidermal cell and then the fungus grow intercellularly. Afterwards the fungus caused cellular collapse and necrosis of cortex parenchyma, necrosis of epidermal and cortex cell, and collapse, necrosis and cellular disintegration of pith in roots of chickpea and tissues degeneration which accelerated the death of host. The present findings are in confirmation with the work of Netzahualcoyotl Mayek - Perez *et al.* (2002).

In physiological parameters where chickpea genotypes (Heat tolerant variety trial (Desi), HVT Large seeded, National nursery dry root rot, Confirmation dry root rot) were evaluated for Relative water content and chlorophyll content. The studies led to conclusion that maximum water holding capacity and higher chlorophyll content developed resistance in chickpea genotypes namely JG 16 and ICC 16181 having maximum Relative water content and maximum chlorophyll content (SPAD unit). In HVT (Large seeded) ICCV 07118 followed by JG 14 x 4958, JG 130 x ICC 11551 and ICCV 11114 which posses maximum Relative water content and chlorophyll content has less percent mortality due to dry root rot and considered as resistant. It is evident from the data summarized that maximum water holding capacity in mesophyll cell of leaves and maximum chlorophyll content of leaves might have played a major role in making the plant resistant to disease. Schlemmer *et al.* (2005) observed that

physiological parameters provide information for interpreting its response to the environment. This study examined the relationship of chickpea leaf spectral response to its chlorophyll content, relative water content and effects of N stress and water stress. Dry root rot disease is directly correlated to moisture stress and high temperature environment.

In the present investigation higher incidence of dry root rot in chickpea varieties under higher temperature conditions revealed that the pathogen infected chickpea plants and manifested dry root rot faster at high temperature $> 33^{\circ}\text{C}$. as the high temperature prolonged plant becomes more vulnerable to this disease. Hence, present studies clearly demonstrate the combined roles of high temperature and the pathogen that infects chickpea and causes the development of dry root rot. Sharma and Pande (2013) indicated that drought caused by prolonged high temperatures make chickpea more vulnerable to dry root rot caused by the fungus *Rhizoctonia bataticola*.

SUMMARY, CONCLUSIONS AND SUGGESTIONS FOR FURTHER WORK

Dry root rot of chickpea caused by *Rhizoctonia bataticola* is an important disease which varies depending upon the inoculum, physical condition of the soil and weather condition. The disease has been observed at flowering / podding stage of the crop. Disease plants exhibited straw colour appearance with blackened roots which later on become brittle. Dark minute, sclerotial bodies were observed on the roots exposed or inside the wood.

Screening of disease sick field led to conclusion that among 50 entries / genotypes, 17 entries found to be resistant to dry root rot. The disease ranged from 2.7 to 100 percent. Further, under blotter paper technique 5 genotypes were found resistant.

Histopathological observation of chickpea dry root rot on the interaction of host and pathogen in infected plants showed typical symptoms on the root surface. Sclerotia comes in contact with the chickpea root surface and developed hyphae which penetrate in tissues of root and then grow intercellularly and fungus caused cellular collapse and necrosis of epidermis, cortex, pith, and xylem and phloem vessels.

Studies on physiological parameters in relation to Relative water content and chlorophyll content with disease incidence concluded that the effect of nitrogen stress and water stress in plant lead to development of dry root rot under high temperature environment. The genotypes having maximum Relative water content and Chlorophyll content under the same environment were found to be resistant.

Conclusion

Dry root rot of chickpea caused by *Rhizoctonia bataticola* is becoming a serious problem when the ambient day temperature is exceeding $> 30^{\circ}\text{C}$ during flowering and pod formation stage. Studies led to conclusion that chickpea genotypes under high temperature environment possess resistance which have high relative water content and chlorophyll content thus, indicated their possible role in development of resistance in genotypes. The entries exhibited resistance under *in vitro* as well as under sick field may be utilized for resistance breeding programme after further confirmation for their level of resistance under field condition.

Suggestions for further work

The entries / genotypes which were found resistant against dry root rot (*Rhizoctonia bataticola*) under *in vitro* and field condition needs to be tested in natural field condition in infested soil under multilocation evaluation test.

Genotypes coupled with high relative water content (RWC %), chlorophyll content, resistance to dry root rot needs further confirmation to incorporate these characters in high yielding cultivars of chickpea.

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Table 1. Meterological data during the cropping season (October 2012 – April 2013)

Month	Week No.	Temperature(⁰ C)		Rain Fall(mm)	Sunshine hours	Relative Humidity	
		Mean Min.	Mean Max.			Min.	Max.
October	40	33.4	21.9	5.2	9	81	44
	41	32.4	18.5	0	8.3	83	35
	42	32.5	18.6	0	9.3	79	32
	43	21.6	15.2	0	8.9	82	35
	44	27.9	15.6	0	9	91	51
November	45	28.2	14.6	0	8.3	88	30
	46	28.6	10	0	8	85	28
	47	27.9	10.6	0	8.7	88	29
	48	28.4	11.5	0	7.3	83	33
December	49	28.7	10.6	0	8.1	85	26
	50	29	14	0	8.2	92	41
	51	25.3	7.1	0	8.6	88	29
	52	23.8	5	0	7.4	90	30
January	1	23.3	7.2	20.6	5.8	87	42
	2	23	5.2	0	7.1	87	32
	3	26.7	10.1	0	8.6	84	36
	4	21.4	5	13.2	6.8	86	36
	5	24.6	7.4	7.6	4.5	91	36
February	6	25.9	11.3	0	8.6	88	49
	7	25.2	13	0	8.3	91	60
	8	25	11	0	9.6	93	49
	9	28	9.2	0	9.1	87	37
March	10	30.7	9.9	0	8.3	83	28
	11	31.6	14.7	0	7.8	84	39
	12	33.4	16.2	0	8.7	77	28
April	13	33.8	16.2	0	8.5	76	30
	14	35.3	15.2	3	8.3	68	20
	15	39.5	19.9	0	8.2	57	14
	16	36.5	20.1	0	8.5	51	22

Vita

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